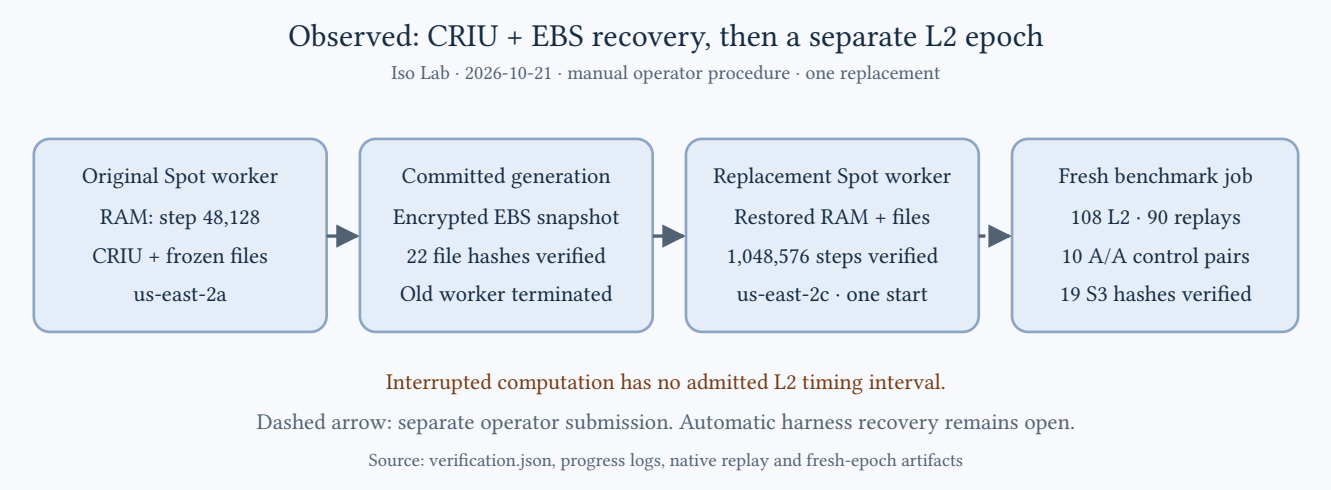
CRIU + EBS recovery and a fresh L2 epoch

Iso Lab · 21 October 2026 · operator validation

Observed: a native SHA-256 process resumed from RAM and an encrypted EBS snapshot on a replacement EKS Spot worker. Native replay verified its complete output. A separately submitted fresh benchmark epoch passed the existing L2, replay, A/A and S3 checks. **Automatic recovery of ordinary harness jobs remains unimplemented.**



Check	Observed result
Process checkpoint	Step 48128 CRIU 4.2.1; UID/GID 1000
Replacement	c8a.2xlarge; us-east-2a ⟶ us-east-2c
Original writer fence	EC2 terminated at 05:17:19.212 UTC; replacement pod created at 05:17:20 UTC
Checkpoint consistency	22 file hashes; encrypted complete EBS snapshot
Native process replay	1048576 final steps; 1024 recorded values; 1 process start
Fresh epoch	108 L2 executions; 90 reproduced measured replays
A/A controls	10 pairs admitted; recorded 95% ratio interval 0.989244–1.012734
S3 verification	19 payload hashes; complete manifest
Fresh worker	Eight logical CPUs; eight physical cores; no SMT

The SHA-256 chain is a correctness fixture, not a speed benchmark. Its pauses allow checkpointing. The fresh ecbench job uses the registered 20-bit fixture, two targets, nine arms, five measured rounds and a warmup. No timing crosses the restore boundary. The A/A interval is a recorded control result, not a speedup. Canonical algorithm graphs and leaderboards are unchanged.

Identity and evidence

The running source is retained as `fixture-source-executed.rs`. Its hash is `42b3b70c1f281e1ffe507e653022bd170e9197b95c4e2cb75abd7a7df7193d75`. The running binary hash is `b3cd5b6832a85967fddc5489984963939bd26e76798280a64f0e03b45b4352b8`. The native verifier recomputes every recorded hash from the public seed and checks the exact target and original output prefix.

Process outcome: `fdde49b4c5a6b75e9974a8993090b4abb7bca85454a0466eece1357b792dae98`. Old instance: `i-0fabdc1b9d167d430` replacement: `i-0ff791a8db99aac86`. Fresh job: `criu-resume-20261021-epoch-1`. Fresh pod UID: `d5afd859-72e2-4288-a0ca-a85b493f58bf`. Fresh source: `cb2ec6873f30b683f7885b53c410d69965f02c20`.

Full details, exact commands, receipt identities, source hashes and private S3 locations are in `REPORT.md`, `protocol.json`, `fresh-epoch-protocol.json` and `verification.json`. Raw process memory stays in encrypted private storage and is excluded from this report/Git.

Failures and limits

The initial process start lacked `setpriv` and its dump found no process. The first archive lacked `tar` and produced an empty S3 object, rejected before generation commit. Both failures are retained. Corrected tool setup and download verification produced the committed generation. A transient replacement Spot quota/capacity error cleared without raising the 96-vCPU quota or 64-vCPU lab cap.

Containerd CRI restore-via-create stayed disabled. The privileged operator supervisor restored a trusted child; ordinary experiment pods retain Restricted admission. This one test does not qualify arbitrary commands, perf/network/GPU state, other instance families or unattended recovery. Its fresh epoch was an operator submission and is not the pending PR 2342 main-branch Action proof.

The remaining controller/API and benchmark epoch integration is tracked in `problems/eks-resumable-l2-experiments.md`. Disposable pods, workers and PVCs were removed; the encrypted snapshot and private S3 evidence are retained.